

STUDYNEST CHROMA

Vivid Academic Syllabus, Engineering Study Hub & File Management Portal

A modern, responsive React 19 + Vite 8 web application providing instant unit-wise PDF study notes, accredited textbooks, NBA CO-PO correlation matrices, credit-weighted SGPA estimation, virtual lab simulators, and real-time administrative file management.

LIVE PRODUCTION URL: <https://krishnajith17.github.io/studynest/>
GITHUB REPOSITORY: <https://github.com/krishnajith17/studynest>

PROJECT AT A GLANCE — KEY METRICS

11 Courses

Sem 1 & 2 Accredited EAC Subjects

33+ Modules

Unit Breakdowns & Instant PDFs

50+ Books

Textbooks & References with ISBN

5 Palettes

Vivid Themes + Dark/Light Mode

CORE ARCHITECTURAL PILLARS

1. Unified Academic Hub

Complete syllabus objectives, unit topics, lab experiments, CO-PO mapping matrices, and textbook links in one interface.

2. Client-Side PDF Engine

Instant browser-based generation of formatted course syllabus PDFs and unit-specific handouts using jsPDF.

3. Real-Time Admin File Manager

Passcode-protected Admin section to Upload, Remove, and Exchange (swap) study documents in real time.

4. Mobile & Laptop Responsive

App-style fixed bottom navigation bar on mobile phones (iOS/Android) and multi-column glassmorphic layout on laptops.

PROJECT PRESENTATION METADATA

Developer & Presenter: Krishnajith (krishnajith17)

Program: B.Tech Electronics & Computer Engineering (EAC)

Admin Passcode: STUDY | Version: 3.0.0

Date: September 29, 2026

1. PROBLEM STATEMENT & PROJECT MOTIVATION

1.1 Challenges Faced by Engineering Students & Faculty

- **Fragmented Curriculum Documents**
Official university syllabus books are static 150+ page PDFs. Locating a single unit topic, textbook ISBN, or lab experiment list requires scrolling through dozens of pages.
- **Lack of Quick Unit-Level Handouts**
Students frequently need a clean, printable summary of a single module (e.g., Unit 2 of Digital Electronics) rather than an entire multi-semester document.
- **Static & Hard-to-Update Study Materials**
When faculty or class representatives update lecture notes or question banks, sharing links across chat groups leads to version confusion and broken files.
- **Manual SGPA & Grade Estimation**
Calculating credit-weighted SGPA across 1-credit labs, 3-credit theory courses, and 4-credit core subjects using the Amrita grading scale (O, A+, A, B+, B, C, P) is error-prone.
- **Poor Mobile Experience on Academic Portals**
Traditional syllabus tables break on smartphone screens and lack touch-friendly navigation for students studying on the go.

2. PROJECT OBJECTIVES & SOLUTION SCOPE

2.1 How StudyNest Chroma Solves These Challenges

- **Interactive Single-Page Curriculum Hub**
Consolidates all 11 Semester 1 & 2 B.Tech EAC courses with instant search, category filtering (SCI, ENGG, HUM), and credit sorting.
- **Automated Client-Side PDF Generator**
Compiles official multi-page PDF syllabus documents or targeted single-unit study guides on the fly in under 1.5 seconds.
- **Dynamic Admin File Manager (Upload, Remove, Exchange)**
Enables administrators to upload custom PDFs/notes, remove outdated handouts, or exchange (swap) a file in-place while preserving all course links.
- **Built-In Engineering Simulators & Calculators**
Provides an interactive SGPA/CGPA calculator, Ohm's Law & RC filter solver, Digital Logic truth-table simulator, and Viva-Voce Q&A bank.
- **Chromatic Multi-Palette UI & Mobile App Bar**
Features 5 vivid color palettes with Dark/Light modes and a dedicated iOS/Android bottom navigation bar with 44px touch targets.

3. TECHNOLOGY STACK & ENGINEERING SPECIFICATIONS

Layer / Domain	Technology Used	Role & Technical Justification
Frontend UI Library	React 19.2.8	Component-driven SPA architecture using Hooks (useState, useMemo, useCallback)
Build & Bundler	Vite 8.3.0	Sub-500ms production rollup bundling + custom dev-mode HTML transform plugin
Styling & Themes	Custom CSS3 Tokens	3-layer CSS variable architecture supporting 5 palettes + Dark/Light mode
PDF Generation	jsPDF 4.2.1	Client-side vector PDF creation with dynamic pagination, headers, and tables
Icon System	Lucide React 1.31	Lightweight, consistent SVG iconography across cards, drawers, and modals
State Persistence	Browser localStorage	Zero-backend persistence for custom courses, Base64 files, bookmarks & history
CI/CD & Hosting	GitHub Pages + Actions	Automated deployment via deploy.yml + root/docs static asset synchronization

4. APPLICATION ARCHITECTURE & COMPONENT HIERARCHY

- **App.jsx (Core Controller & State Router)**

Manages global state via useLocalStorage ('studynest_chroma_courses', 'bookmarks', 'history', 'theme', 'palette') and routes between Browse, SGPA Calculator, Learning Hub, and Admin views.

- **Navbar.jsx & ThemePalettePicker.jsx**

Provides desktop top navigation, quick counters for Bookmarks and History, real-time 5-palette theme switcher, and the fixed mobile bottom navigation bar.

- **HeroSection.jsx & FilterBar.jsx**

Displays live curriculum metrics (Courses, Modules, Textbooks), real-time search input, Semester pills (Sem 1 / Sem 2), Category tabs, and sorting dropdown.

- **CourseCard.jsx & CourseDetailModal.jsx**

Renders individual subject cards with category aura glows and opens the 6-tab deep-dive modal (Syllabus, CO-PO Matrix, Labs, Textbooks, Simulators, Study Files).

- **AdminPanel.jsx & FileExchangeModal.jsx**

Handles passcode-protected curriculum administration, new course creation, and the complete File Management suite (Upload, Remove, Exchange/Swap).

- **DownloadModal.jsx & pdfGenerator.js**

Executes a 3-stage verification and compilation progress flow, serving either custom faculty-uploaded files or dynamically generated jsPDF documents.

5. STUDENT PORTAL — CORE FUNCTIONAL MODULES

5.1 Intelligent Search, Filtering & Sorting Engine

- **Multi-Field Instant Search**
Queries course codes (e.g., '23ECE102'), course titles, descriptions, syllabus unit names, and textbook titles/authors simultaneously with zero latency.
- **Category & Semester Segmentation**
One-click filtering across Semester 1, Semester 2, Sciences & Math (SCI), Core Engineering (ENGG), and Humanities (HUM).
- **Multi-Criteria Sorting**
Sort courses by Course Code, Alphabetical Title, Credits (High-to-Low), or Semester progression.

5.2 Inside the 6-Tab Interactive Course Detail Modal

- **Tab 1 — Syllabus & Units**
Displays Course Educational Objectives and expandable Unit breakdowns. Students can check off completed units to track exam preparation and download unit-specific PDFs.
- **Tab 2 — Course Outcomes & NBA CO-PO Matrix**
Lists measurable Course Outcomes (CO1–CO5) alongside a full 15-column NBA Program Outcome (PO1–PO12) and Program Specific Outcome (PSO1–PSO3) correlation table.
- **Tab 3 — Laboratory Experiments**
Available on lab-integrated courses (e.g., 23ECE101, 23ECE102, 23EEE184, 23CSE101), detailing hardware/software experiments and prototype evaluation requirements.
- **Tab 4 — Prescribed Textbooks & Reference Books**
Complete bibliographic records including Author, Edition, Publisher, Publication Year, ISBN, and 1-click external search links to Google Books.
- **Tab 5 — Virtual Simulators & Video Lectures**
Direct integration with interactive engineering tools including Falstad Circuit Simulator, GeoGebra 3D, Desmos Graphing Calculator, Python Tutor, and NPTEL courses.
- **Tab 6 — Official Study Files & Handouts**
Dynamically appears when files are uploaded via the Admin Panel, allowing students to download custom lecture notes, question banks, and formula sheets.

5.3 Personalization: Pinned Bookmarks & Download History

- **Bookmarks Slide-Over Drawer**
Students can star/pin important courses for instant 1-click access from the top navigation bar or mobile bottom bar.
- **Recent Downloads Drawer**
Automatically logs the last 30 downloaded syllabus PDFs or unit handouts with timestamps and a 1-click 'Re-Download' button.

6. INTERACTIVE SGPA CALCULATOR & LEARNING HUB

6.1 Credit-Weighted SGPA & CGPA Estimator (SgpaCalculator.jsx)

- **Accredited Grading Scale**
Implements the official Amrita 10-point grading system: O (10.0), A+ (9.5), A (9.0), B+ (8.0), B (7.0), C (6.0), P (5.0), and F (0.0).
- **Exact Credit Weighting**
Computes $SGPA = \frac{\text{Sum}(\text{Course Credits} \times \text{Grade Points})}{\text{Sum}(\text{Total Credits})}$ across Semester 1 (23 Credits) and Semester 2, plus cumulative CGPA estimation.

6.2 Interactive Engineering Learning Hub (LearningHub.jsx)

- **Circuit & RC Filter Calculator**
Real-time solver for Ohm's Law ($V = I \times R$, $P = V \times I$) and First-Order RC Low-Pass Filter Time Constant ($\tau = R \times C$) and Cutoff Frequency ($f_c = 1 / 2\pi R C$).
- **Digital Logic Gate Simulator**
Interactive binary input switches (A, B) with live boolean output evaluation and truth table highlighting for AND, OR, NAND, NOR, XOR, XNOR, and NOT gates.
- **Viva-Voce Question Bank**
Curated oral exam questions and concise technical answers covering IoT, Digital Logic, Basic Electrical Engineering, C Programming, and Engineering Chemistry.

7. ADMIN PANEL & REAL-TIME FILE MANAGEMENT SYSTEM

7.1 Complete File Lifecycle: Upload, Exchange & Remove

- **Passcode Authentication**
Protected by passcode 'STUDY' via AdminLoginModal.jsx to prevent unauthorized edits.
- **Upload Study Files**
Admins select a target course, assign a module/unit, classify the document (Lecture Notes, Question Bank/PYQ, Lab Manual, Formula Sheet, Syllabus), and upload files up to 5MB.
- **Exchange / Swap Files In-Place (FileExchangeModal.jsx)**
Allows replacing an existing file with an updated version while preserving the course mapping, unit association, and student download buttons. Displays old vs. new file size comparison.
- **Remove Files & Storage Quota Tracker**
One-click deletion with confirmation prompt, paired with a live progress bar monitoring total KB/MB used out of the 5.0 MB browser storage quota.
- **Dynamic Course Builder & Factory Reset**
Admins can create new courses for Semesters 1–8, add custom syllabus units, or reset the entire database back to factory defaults.

8. ACCREDITED B.TECH EAC CURRICULUM DATABASE (11 COURSES)

Code	Sem	Cat	Cr	L-T-P	Course Title & Primary Prescribed Textbook
23MAT124	1	SCI	4	3-1-0	Calculus & Matrix Algebra — Strang (Linear Algebra), Anton (Calculus)
23ECE101	1	ENGG	3	2-0-3	Introduction to Internet of Things — Bahga & Madiseti (IoT: A Hands-On Approach)
23ECE102	1	ENGG	4	3-0-3	Digital Electronics & Systems — Morris Mano (Digital Design), Floyd
23EEE104	1	ENGG	3	3-0-0	Basic Electrical & Electronics Engg. — Alexander & Sadiku, Boylestad
23EEE184	1	ENGG	1	0-0-3	Basic Electrical & Electronics Lab — Kothari & Nagrath, Floyd
23CSE101	1	ENGG	4	3-0-3	Computational Thinking & Problem Solving — Ferragina, Gutttag (MIT Press)
23ENG101	1	HUM	3	2-0-3	Technical Communication — Raman & Sharma (Oxford), Markel
23CUL101	1	HUM	2	2-0-0	Cultural Education I — Amrita University Publication, Radhakrishnan
23ECE103	2	ENGG	4	3-1-0	Network Analysis & Synthesis — Hayt, Kemmerly & Durbin, Van Valkenburg
23ECE104	2	ENGG	4	3-0-3	Electronic Devices & Circuits — Sedra & Smith (Microelectronic Circuits)
23CHY104	2	SCI	3	2-0-3	Engineering Chemistry — Jain & Jain (Dhanpat Rai), Atkins (Physical Chem)

8.1 Department Framework & NBA Accreditation Alignment

- Program Educational Objectives (PEO1–PEO3)**

Prepares graduates for hardware-software co-design careers, embedded & cyber-physical systems innovation, and lifelong ethical engineering leadership.

- Program Specific Outcomes (PSO1–PSO3)**

Covers solid-state electronic devices, signal processing, IoT sensor networks, and modern algorithmic software engineering.

9. CHROMATIC DESIGN SYSTEM & CROSS-DEVICE RESPONSIVENESS

9.1 Five Real-Time Dynamic Theme Palettes

- **1. Electric Aurora (Default)**
Deep Indigo surface (#080c18) paired with Cyber Cyan (#00f0ff), Electric Purple (#8b5cf6), and Neon Mint (#10b981).
- **2. Sunset Radiance**
Obsidian Plum surface (#120b14) paired with Solar Coral (#ff5722), Amber Gold (#f59e0b), and Sunset Fuchsia (#ec4899).
- **3. Cyberpunk Neon**
High-contrast Onyx (#05050a) paired with Laser Rose (#f43f5e), Hyper Cyan (#06b6d4), and Acid Lime (#a3e635).
- **4. Emerald Oasis**
Abyssal Forest (#04120e) paired with Radiant Emerald (#10b981), Aqua Mint (#2dd4bf), and Spring Lime (#84cc16).
- **5. Cosmic Amethyst**
Midnight Violet (#0d071a) paired with Galactic Orchid (#c084fc), Vivid Fuchsia (#d946ef), and Starlight Gold (#facc15).

9.2 Step-by-Step Classroom Live Presentation Script (5-Minute Walkthrough)

- **Step 1 — Introduction & Theme Showcase (1 Min)**
Open <https://krishnajith17.github.io/studynest/> on the classroom projector. Highlight the instant dark load, curriculum stats, and switch live between the 5 color palettes and Light/Dark modes.
- **Step 2 — Course Exploration & Instant PDF Generation (1.5 Min)**
Search for 'Digital' or filter by Semester 1. Open '23ECE102 Digital Electronics', showcase the Unit syllabus, NBA CO-PO matrix, Accredited Textbooks, and click 'Download Course Syllabus PDF' to generate a live PDF.
- **Step 3 — SGPA Calculator & Learning Hub Simulators (1 Min)**
Switch to the SGPA Calculator to compute semester GPA live. Then open the Learning Hub and toggle inputs on the Digital Logic Gate Simulator and RC Filter Calculator.
- **Step 4 — Admin File Manager: Upload, Exchange & Remove (1.5 Min)**
Click 'Admin' (Passcode: STUDY). Open the File Manager tab, upload a sample PDF handout to a course, demonstrate the 'Exchange' button to swap it with a revised file, and show it appearing immediately in the student Course Modal.